

A Basketball Shot Probability Model

J. Remmert & D. Sharkansky

Uni ID: jcr2232, dss2204

1. Introduction

This project develops a Bayesian expected shot model for basketball. While shot probability models are not new, nearly all public versions rely on frequentist methods such as Logistic Regression, XGBoost, or Neural Networks. These approaches typically use a shot's x and y coordinates, transform them into polar coordinates relative to the hoop, and estimate the probability as a function of distance and angle. Most existing models also focus primarily on shot location and rarely incorporate shooter or defender identity. After estimating shot probability, these models compute an expected value metric by comparing the predicted probability to the observed outcome. They interpret the difference as the added (or lost) value by the player, a framework that mirrors expected goals in soccer and expected batting metrics in baseball. A shortcoming with the frequentist framework is that it provides only point estimates. To address this limitation, we propose a Bayesian approach that not only directly estimates player and defender effects but also quantifies uncertainty.

The goal of this work is to separate shot difficulty into its location, shot type, player skill, defensive skill, and court context. Our model is in two stages. We first estimate the likelihood of a made shot given its distance and angle via Bayesian Additive Regression Trees (BART). This is what we refer to as the baseline probability. BART is a non-parametric Bayesian approach that is akin to XGBoost. BART sums over shallow trees fit to the previous tree's residual, with each tree contributing a small amount. The Bayesian aspect is we implement a prior over the tree parameters and use MCMC to fit. To ensure a calibrated BART model, we also trained a Logistic Regression and XGBoost model. For the second stage, we scale the baseline probability and layer in hierarchical effects for shot type, player, defensive team, and court information. Conceptually, the hierarchical effects soak up the residual and identify the player's latent skill for a shot or a team's defensive ability. For example, the empirical probability to score a dunk from anywhere on the court is 88% versus 39% for a jump shot. If a player continually makes a dunk or misses a jump shot, their player effect will be positive or negative respectively, which directly translates into their skill. Through hierarchical modeling, we leverage partial pooling, which shrinks individual effects towards a population mean given their proportion in the dataset. For example, if we observe many shots from Stephen Curry, the model can identify his true talent relative to the mean. However, for a generic role player with limited attempts, no matter their success rate, the model will shrink their effect closer to the mean. We extend this pooling framework across seasons. Rather than treating player skill as independent for each year, we decompose performance into a player skill by shot type and model season-level deviations through a mean-centered random walk. The player skill captures their talent, while the walk captures fluctuations around it. To summarize, we estimate the likelihood of a shot given its location, then use the mixed effects to shift the baseline value up or down. We attribute this difference to the features, which allows us to estimate shot type difficulty, player skill, and defensive prowess.

2. Data and Features

Prior to establishing our models, we provide a short description of our data, which we collected from the NBA’s API with Python. This dataset is not as detailed as desired. For example, closest defender or defender distance would allow for a more accurate model. We collected shot data from the 2022/23 to the 2024/25 seasons for a total of 512,943 shots.

Table 1: Description of Features

Variable Name	Type	Description
shotResult	Binary	The target of our model. Indicates if the basket went in or not.
X, Y	Continuous	The shot location in X (width) and Y (length) coordinate space. The origin of our coordinate system was in reference to the hoop for our dataset. These measurements were in tenth of an inch.
sub shot type	Categorical	Detailed descriptor of the shot such as Turnaround Fadeaway Shot, Driving Floating Bank Jump Shot, Tip Layup Shot, etc.
period	Categorical	Which period the shot occurred in: 1, 2, 3, 4, OT.
clock	Continuous	Current game clock in seconds when the shot occurred. This is not the shot clock.
Player ID	Categorical	A unique vector encoding the player IDs.
Defensive Team	Categorical	The opposition the shooter is facing.
Home Team ID	Categorical	A unique vector encoding the Home Team IDs.
Player Height	Continuous	The player’s height as measured in inches.
Player Age	Continuous	The player’s age as measured years.

3. Preprocessing and Engineering

After collecting the data, we removed 5 games and 112 players where the dataset was missing home team or player height. We then aggregated the shot subtype into four categories given their names: Jump Shot, Layup, Dunk, and Hook. Please see Appendix 7 for the exact mapping.

To compute distance, we calculated the euclidean distance in feet between shooter location and hoop. We computed the angle of each shot from the center of the court relative to the basket, mirroring all shots to one side the court. This gives a measure in degrees of how far a shot is from the center, with smaller angles near the middle and larger angles toward the corners. Shots from the left and right sides of the court are treated symmetrically. Formally:

$$\text{distance} = \sqrt{(X/10)^2 + (Y/10)^2}$$

$$\text{angle} = \arctan\left(\frac{|x|}{y}\right) \times \frac{180}{\pi}$$

We removed shots beyond 48 feet (one foot beyond half-court) and beyond 100 degrees (just beyond the corners). Shots from these areas are difficult and rarely attempted. After removal, we have 503,749 shots left. Next, we split the data 70/30 into a train and test set, then scaled distance and angle to be between 0 and 1 for modeling.

Table 2: Engineered Features

Variable Name	Type	Description
shot type	Categorical	Descriptors of the shot such as Jump Shot, Layup, Dunk, or Hook.
shotDistance	Continuous	Euclidean distance between shooter location and hoop. Distances range from 0 (on top of the basket) to 48 feet (one foot beyond half-court). We removed shots beyond this mark because of their rarity.
angle	Continuous	Measured between 0 degrees (straight on) and 100 degrees (just beyond the corners). Also, we mirrored shots on the left and right to be symmetric. Shots do occur beyond 100 degrees, but very sparse and were removed as a result.

4. Models

4.1. Estimating Baseline Probability

To establish baseline probabilities, we examine three models: Logistic Regression, XG Boost model, and BART. We built the Logistic Regression with Stan, the XGBoost model in Python with sklearn, and BART with PyMC. We did not tune the XGBoost, as its purpose was for comparison.

4.1.1. Logistic Regression

Here we chose a generic normal prior centered at zero. Given the amount of data we have, with over 300,000 shots for training, the data should outweigh any prior selection. We fit this using Stan.

$$y_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \beta_0 + \beta_1 \text{Distance}_i + \beta_2 \text{Angle}_i \quad \beta_j \sim N(0, 1)$$

4.1.2. XGBoost

For XGBoost, we fit this out of the box with sklearn using default parameters.

$$\hat{y}_i = \sum_{k=1}^K f_k(\text{Distance}_i, \text{Angle}_i)$$

4.1.3. BART

BART models the outcome as a sum of m trees, where each tree is represented by the function $g_j(x; T_j, M_j)$. Here, T_j denotes the tree depth, while M_j represents the terminal node parameters or the leaf values. Each individual tree contributes a small amount to the total predicted value, with the sum of these providing the final estimate. We tested both the default priors and priors that favored deeper trees (β of 0.75), yet less splitting (α of 0.75) with 20 trees. The more trees, the more complex the model would become, with most literature recommending 20-50 trees. We did not tune further as our objective is to establish baseline probabilities, show BART’s performance

is on par with XGBoost, and analyze mixed effects.

$$y_i = \sum_{j=1}^m g_j(x_i; T_j, M_j)$$

T_j Tree depth prior that defines the structure of the binary tree

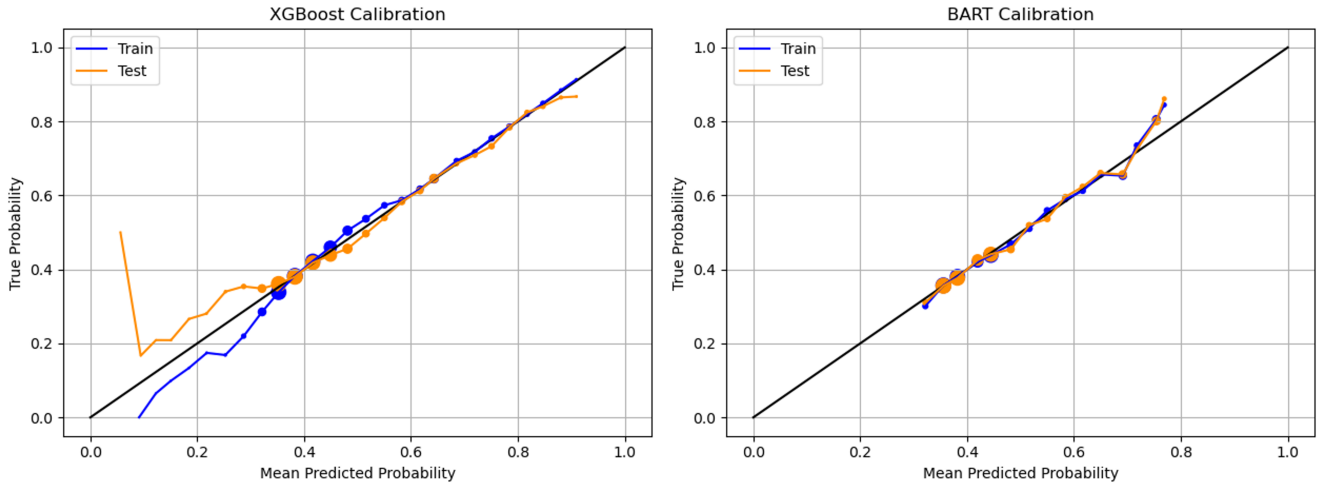
M_j Leaf values prior the represents the terminal nodes

All the models sampled sufficiently given Rhat values were at or below 1.01 and we display the OOS sample performance below. The logistic regression determined posterior weights of -2.1 and -0.04 for distance and angle respectively.

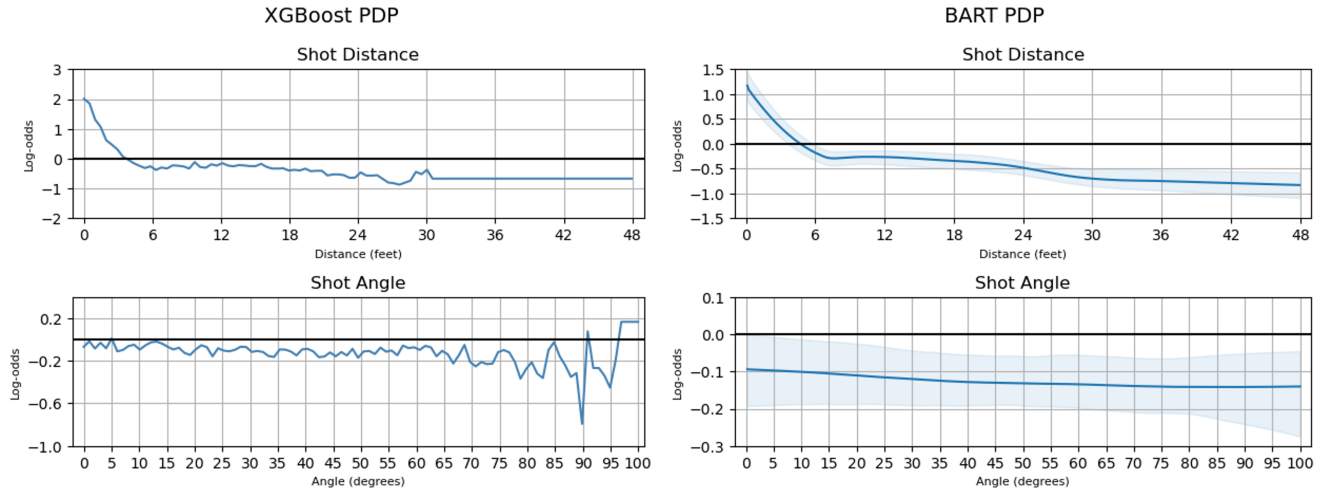
Table 3: OOS Performance

Metric	Logistic	XGBoost	BART	BART with $\alpha = \beta = 0.75$
Train LogLoss	0.6660	0.6460	0.6539	0.6555
Train AUC	0.6372	0.6610	0.6394	0.6384
Train Accuracy	0.5976	0.6253	0.6205	0.6198
Test LogLoss	0.6657	0.6542	0.6538	0.6554
Test AUC	0.6371	0.6370	0.6392	0.6383
Test Accuracy	0.5983	0.6193	0.6209	0.6197

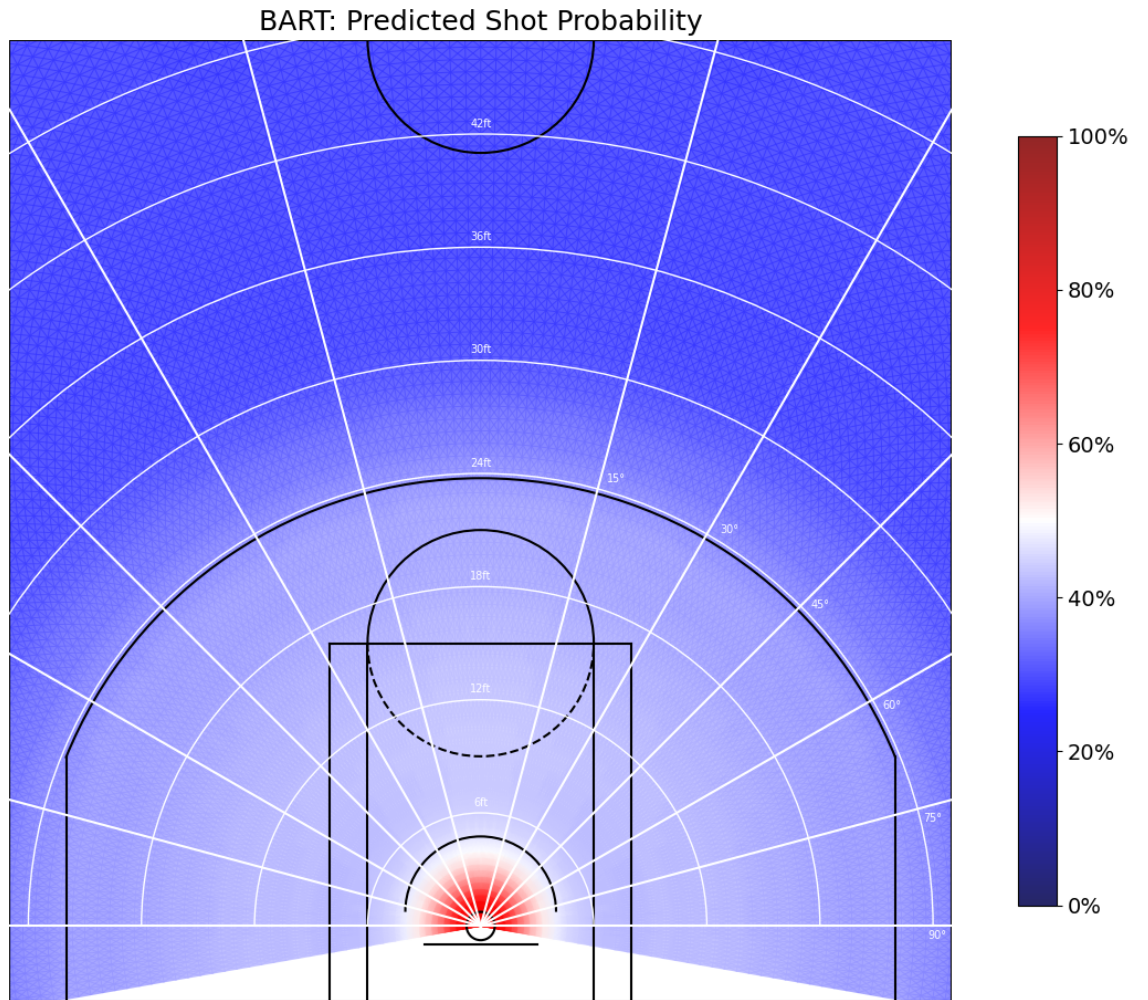
Given the OOS results, BART with default priors performed the best and was well-calibrated relative to XGBoost. This performance was in line with public results 7.2. Please see the BART Appendix 7.3 for convergence checks.



We examined the Partial Dependence Plot for XGBoost and BART to understand how Distance and Angle effect the the log-odds. As one gets further away from the basket, the harder the shot is, which is as expected. Secondly, as one moves away from the center to the corners, shot difficulty increases. The XGBoost Angle PDP plot has jagged lines because we did not tune this model. by introducing regularization one could likely find smoother plots.



Lastly, we visualize our predicted shot probability with BART on the court.



4.2. Adding Hierarchical components

With baseline probabilities established, we introduced hierarchical/multilevel/random effects. These terms are used interchangeably in literature. Rather than present our final model, we iterated from a simple model, first introducing a shot type random effect (Jumper, Layup, Hook,

Dunk), to random walks for player-season shot types. Since we have a multi-level approach, if a player was in the test set but not training, we would not learn a player effect for them. Therefore, we combined the classical training and test sets and examined the ELPD-LOO-CV to assess model fit. For each model, we denoted the baseline probability as bp_i for shot i .

4.2.1. Shot Type Random Effect

In this model, each shot type has their own intercept, γ_{s_i} , which represents the average likelihood of making that shot given the type (Jumper, Layup, Hook, Dunk). Here, we assume shot types come from some shared super-distribution. We use a tight prior centered around 1 for the baseline multiplier, β_1 , to reflect the idea that the location of the shot should require little re-scaling.

$$\begin{aligned} y_i &\sim \text{Bernoulli}(p_i) \\ \text{logit}(p_i) &= \beta_1 bp_i + \gamma_{s_i} \\ \beta_1 &\sim N(1, 0.25) \quad \gamma_s \sim N(0, \sigma^2) \\ s_i &\in \{\text{Jumper, Layup, Hook, Dunk}\} \end{aligned}$$

4.2.2. Shot Type and Player-Shot Type Random Effect

Next, we introduced a player-season shot type random effect, α_{a_i, s_i, t_i} as deviations from the shot type average. Each player a has their own shot type effect s that varies per season t . These deviations can be interpreted as a measure of player skill for that shot type. Each shot type s has its own variance parameter, τ^2 , which governs the spread of player-season deviations for that shot type. Player effects for Layups, for example, are drawn from a different distribution from player effects for Dunks, allowing the model to capture the fact that skill variation may be larger for some shot types than others. A larger τ^2 value implies more variability across players for that shot type. For now, we assume seasons are unrelated, which we address in other models. For the player effect components we fit standard normal priors (or half normals if a variance component) as there should be enough data to shift off the prior, and two-thirds of the data should fall within plus or minus 1 on the log-odds scale. It would be rare to have a player effect at plus or minus 3 as that implies a very low or high added probability to a shot.

$$\begin{aligned} y_i &\sim \text{Bernoulli}(p_i) \\ \text{logit}(p_i) &= \beta_1 bp_i + \gamma_{s_i} + \alpha_{a_i, s_i, t_i} \\ \beta_1 &\sim N(1, 0.25) \quad \gamma_s \sim N(0, \sigma^2) \quad \alpha_{a, s, t} \sim N(0, \tau_s^2) \\ s_i &\in \{\text{Jumper, Layup, Hook, Dunk}\} \quad a_i \in \{1, \dots, A\} \quad t_i \in \{2022, 2023, 2024\} \end{aligned}$$

4.2.3. Adding Defensive Team Shot Type and Court Effects

Similar to above, we introduce a defensive team-season effect, δ_{d_i, s_i, t_i} for each team d across the shot types s and allow them to vary per season t . Once again, this acts as a deviation from the shot type mean. Similar to the player effects, we allow for each defensive shot type to have different spreads of skill. We also incorporate a court effect, ζ_c , to capture court characteristics and soak up residual variation in shooting success not explained by the other effects. Both the defensive and court effects are centered to act as relative differences across teams or courts for model identifiability. For these new components, we also fit them using standard normal priors.

$$\begin{aligned}
y_i &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \beta_1 b p_i + \gamma_{s_i} + \alpha_{a_i, s_i, t_i} + \delta_{d_i, s_i, t_i} + \zeta_{c_i} \\
\beta_1 &\sim N(1, 0.25) \quad \gamma_s \sim N(0, \sigma^2) \quad \alpha_{a, s, t} \sim N(0, \tau_s^2) \\
\delta_{d, s, t} &\sim N(0, \eta_s^2) \quad \zeta_c \sim N(0, \pi^2) \\
s_i &\in \{\text{Jumper, Layup, Hook, Dunk}\} \quad a_i \in \{1, \dots, A\} \\
d_i &\in \{1, \dots, D\} \quad c_i \in \{1, \dots, C\} \quad t_i \in \{2022, 2023, 2024\}
\end{aligned}$$

4.2.4. A Random Walk

The previous models assumed that a player-season effect has limited impact on the next season. However, it seems more likely that the player-season effect is related to an underlying talent level that generally remains stable from year to year. If we consider shot attempts as random events, then we would expect to see some variation in a player's shooting percentage over time. To capture this, we model the player-season-shot type effect as a Gaussian random walk across seasons where deviations are correlated over time. To accomplish this, the random walk, $w_{a, s, t}$ accounts for the drift or deviation across seasons and the player skill $\mu_{s, t}$ is the average over those seasons. Therefore the player effect α becomes $\alpha_{a, s, t} = \mu_{a, s} + w_{a, s, t}$. This allows us to separate a player's true talent from noise over the seasons and estimate their latent skill. We tested out three different sets of priors: first standard normal and half normals, then a normal scaled by 5 (a wide prior), and lastly with exponentials (0.5) for variance terms and a $N(0, 3)$ prior for the player and defense effects. We left the court effect centered at $N(0, 1)$ because these should not provide much of a benefit to a shot, as all courts are created equal with fans that generally support their home team.

$$\begin{aligned}
y_i &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \beta_1 b p_i + \gamma_{s_i} + \alpha_{a_i, s_i, t_i} + \delta_{d_i, s_i, t_i} + \zeta_{c_i} \\
\beta_1 &\sim N(1, 0.25) \quad \gamma_s \sim N(0, \sigma^2) \quad \alpha_{a, s, t} = \mu_{a, s} + w_{a, s, t} \\
\mu_{a, s} &\sim N(0, \tau_\mu^2) \quad w_{a, s, t} \sim N(w_{a, s, t-1}, \tau_s^2) \quad \delta_{d, s, t} \sim N(0, \eta_s^2) \quad \zeta_c \sim N(0, \psi^2) \\
s_i &\in \{\text{Jumper, Layup, Hook, Dunk}\} \\
a_i &\in \{1, \dots, A\} \quad d_i \in \{1, \dots, D\} \quad c_i \in \{1, \dots, C\} \quad t_i \in \{2022, 2023, 2024\}
\end{aligned}$$

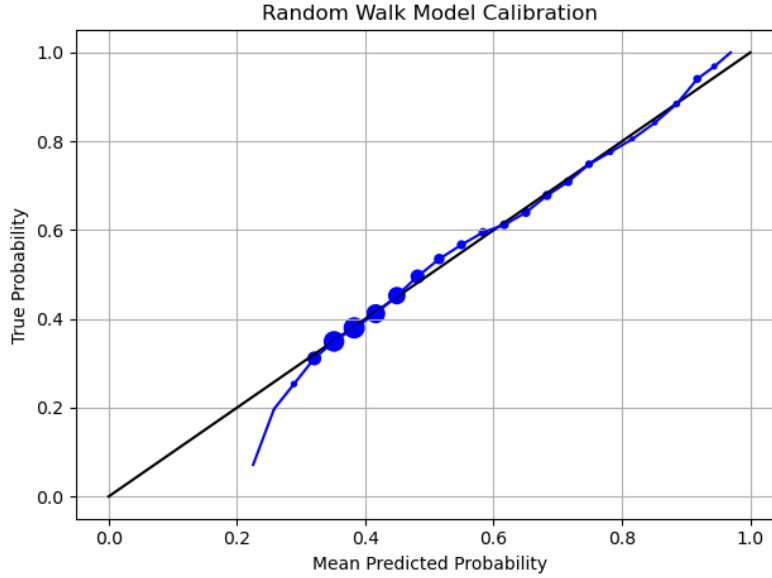
5. Discussion

Rather than compare log loss or AUC, we performed model selection via the ELPD LOO, which indicates the best model is the random walk. Given two models, the one with the higher ELPD LOO has better out of sample performance. Even after manipulating the priors, the best model assumed standard normal and half normal distributions. With all the effects incorporated into the model, we achieved better performance than on BART alone. While we ran the random walk with different priors, we only discuss the most accurate random walk when comparing models given their similar performance.

Table 4: Model Comparison

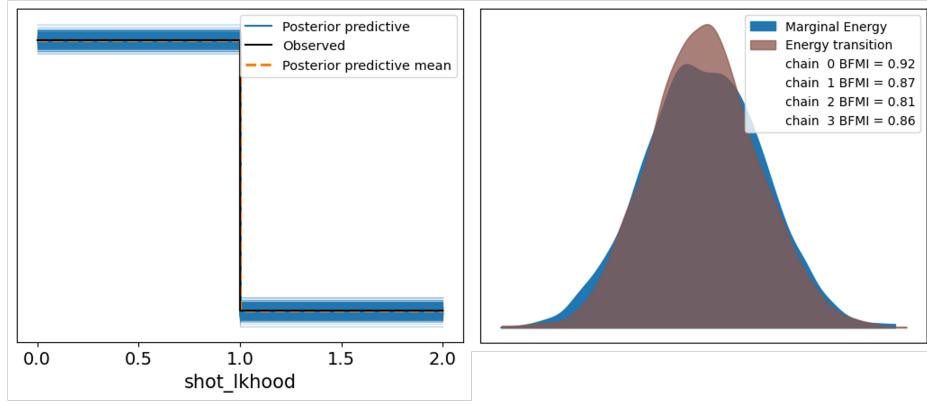
Model	LOO	LOO SE	Log Loss	AUC	Accuracy
Random Walk	-319626.9	205.95	0.6405	0.6587	62.84%
Random Walk, Exp Priors	-319627.6	205.94	0.6405	0.6587	62.84%
Random Walk, Wide Priors	-319628.2	205.96	0.6405	0.6587	62.84%
Def & Court Effect	-319901.1	204.99	0.6401	0.6598	62.88%
Player Effect	-320288.8	203.52	0.6412	0.6582	62.66%
Shot Type	-321218.9	199.82	0.6463	0.6454	62.21%

Further, with the addition of the mixed effect components, our model is well calibrated for the majority of our observations, with the exception of overestimated low-probability shots due to BART’s original conservatism.

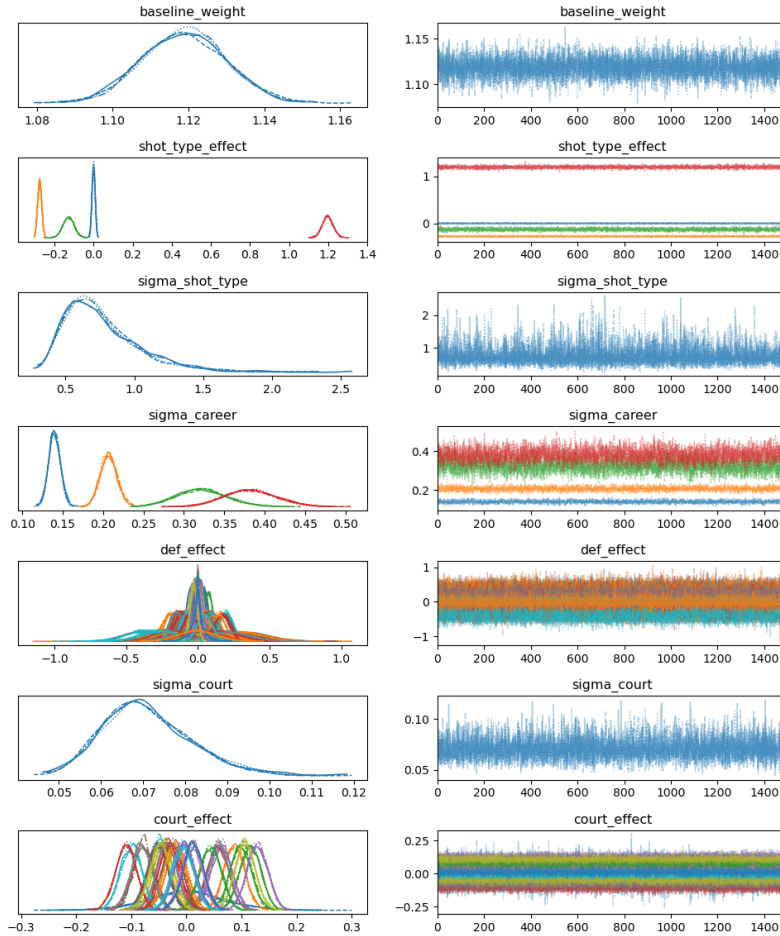


5.1. Diagnostic and Convergence

As discussed earlier, BART converged and its results can be found in 7.3. Here we display some convergence plots for some of our parameters from the random walk model. Each model converged, as implied by their large effective sample sizes and Rhat values below 1.02. The other models had similar plots and we only show them for our best model, the Random Walk. We performed posterior predictive checks and examined how our sampler explored the target distribution. Here we want the posterior predictive to align with the observed data, indicating that our posterior is accurately drawing from our proposed target distribution, and a high degree of overlap implying we have explored the entire proposed target distribution.



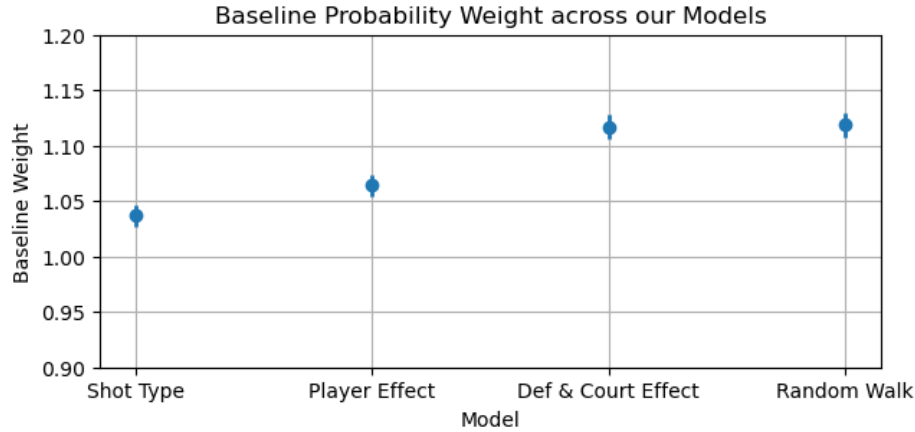
Next we inspected the Rhat and ESS values for each parameter. We found all Rhats were at or below 1.01 as desired and ESS had a minimum of 820, which shows a $820/4000 = 20.5\%$ relative sampling efficiency. We then visually inspected some of the parameters to confirm convergence.



5.2. Baseline Weight

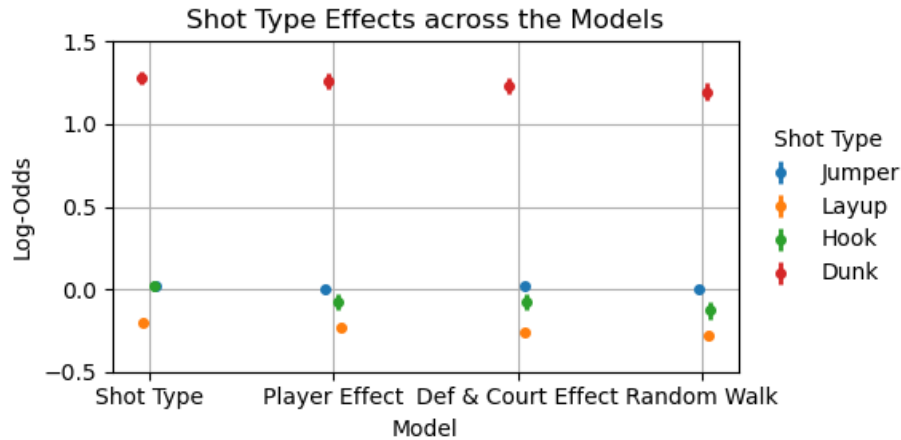
We first examined how the baseline weight changed with the different mixed effect components. As indicated in BART's calibration, the model was conservative, predicting between 20% to 80%, while XGBoost predicted the full range between 0 and 1. BART's conservatism was reflected in the baseline weight being above 1. If we calibrated BART perfectly, it would have a weight of 1. We interpret this as for every 1 unit increase in the log-odds, we scale it slightly for the mixed-effect

models. Please see the appendix for a visual comparison of convergence for the weight across our models, 7.4.



5.3. Shot Type Effect

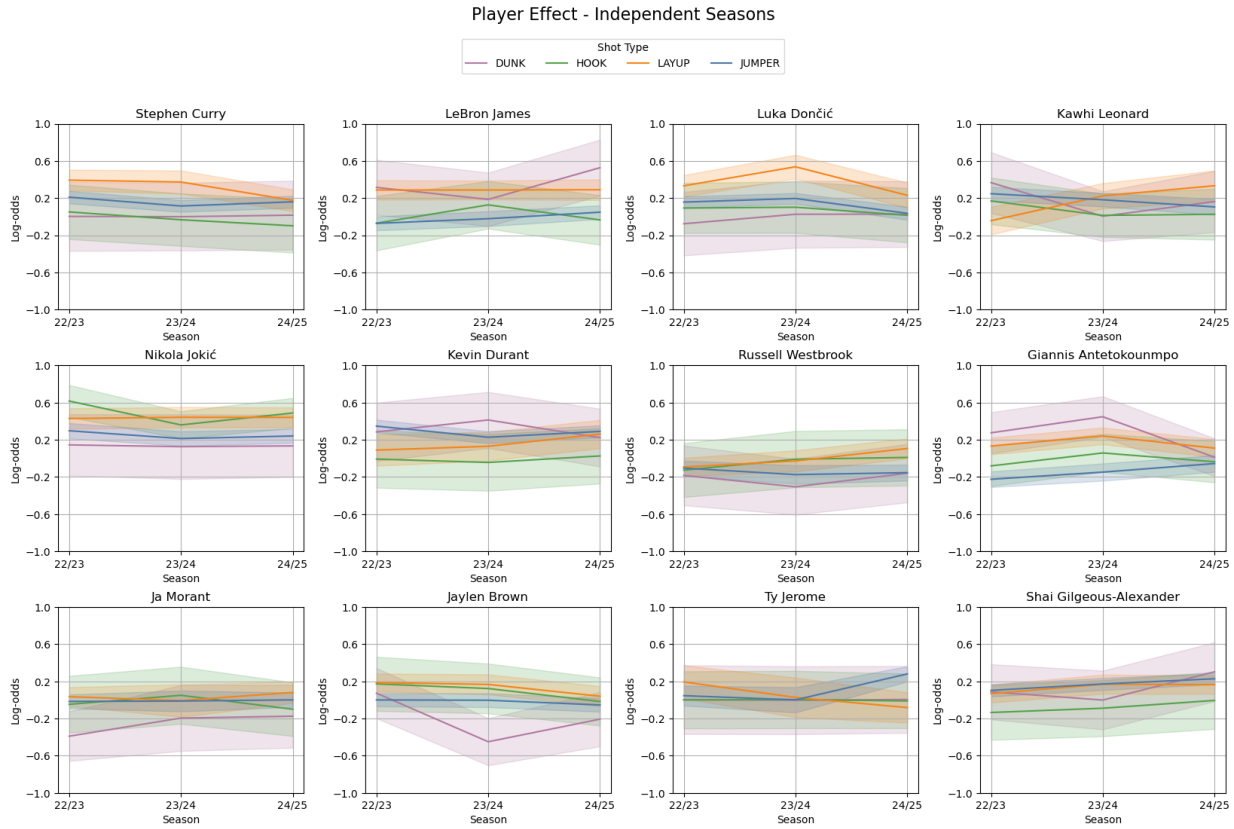
Here we examine the shot type effects. This value represents the average probability for a shot type after accounting for its location on the court. Most notably, we see that BART underestimates dunks and our hierarchical models correct for it, by increasing their baseline odds tremendously. Jumpers do not shift the log-odds too much because they make up majority of our data. Interestingly, we see the baseline odds were overconfident on layups, likely because these are often contested near the hoop. The BART model, considering only distance and angle, saw dunks and layups to be similar and assigned a high likelihood of conversion given their distance to the hoop. The hierarchical model was able to more accurately predict after accounting for shot type. Please see the appendix for a visual comparison of convergence for the shot type effect across our models, 7.4.



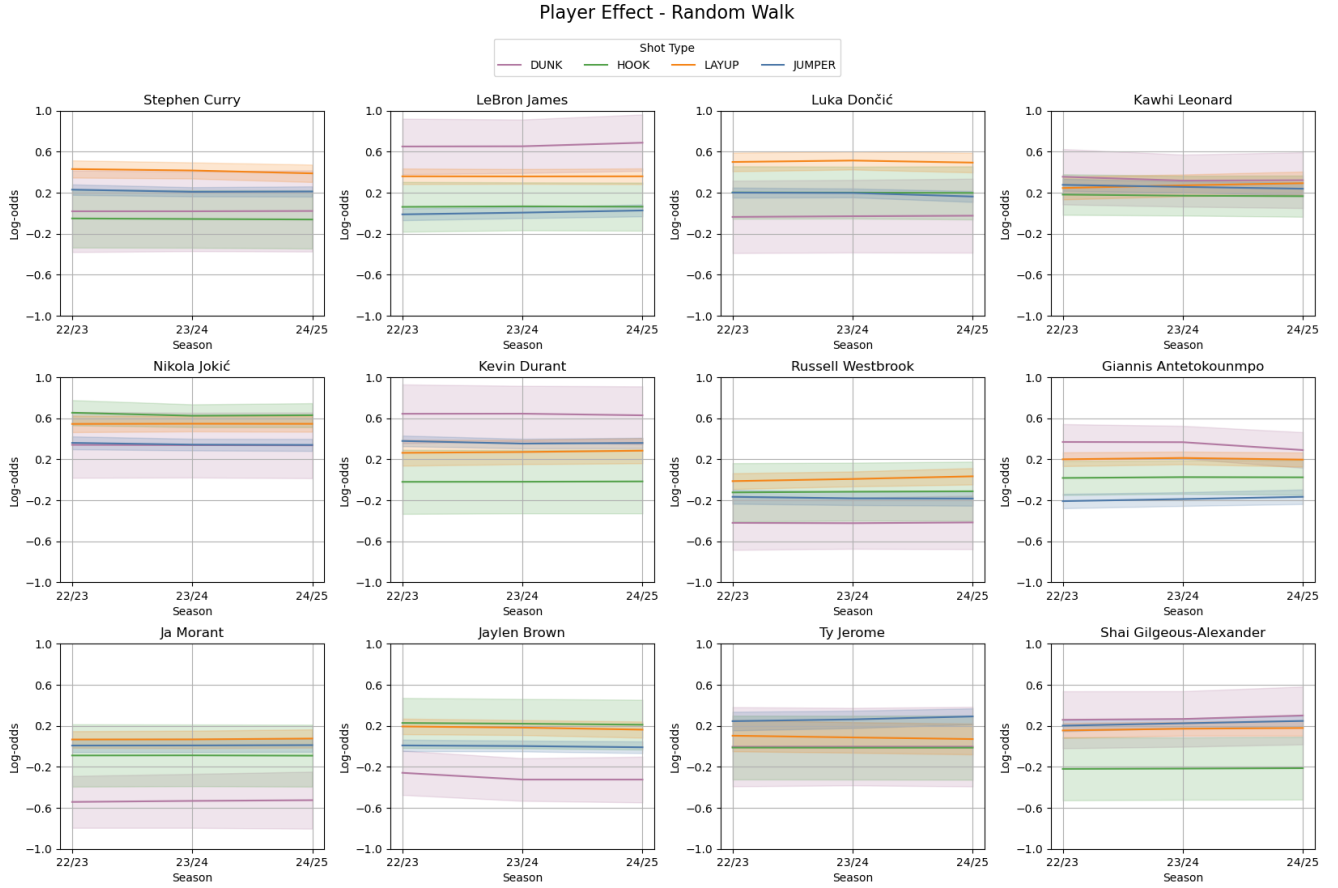
5.4. Player Effect

Here we discuss the player effects over each season, for several players, mostly popular stars. First, we consider the plots below showing player effects without sharing information between seasons from the Defense and Court Model. We can consider these trends as the change in player skill over time. Several of these players appear to have large fluctuations in their shooting ability

for some shot types, particularly in dunks and layups. For example, Luka Doncic shows significant improvement for layups in the 23/24 season before falling below his 22/23 level in the 24/25 season. On the other hand, Jaylen Brown has nearly the opposite trend with dunks.

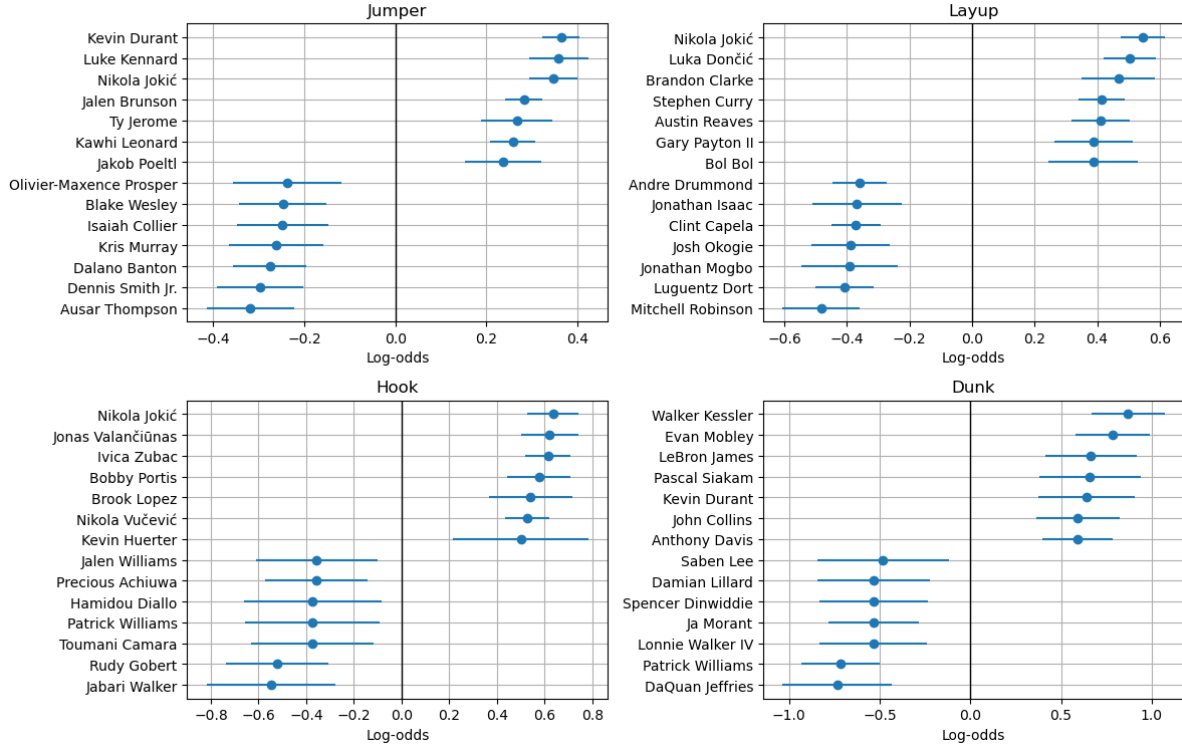


Next we have the player effect with the random walk. As we can see, the trend lines have mostly flattened out. This seems to be a better representation of player skill over time. However, it is hard to tell if the model is capable of acknowledging significant changes in a player's true ability. As an example, when Ty Jerome's season effects are considered independently, there is a large increase in his effect in 24/25, but this is greatly reduced in the random walk model. We suspect the lack of variation is primarily due to the difficulty of estimating a true trend over a limited number of years. If we were to further explore this analysis, getting data for more seasons would be one of our top priorities. Out of curiosity, we also inspected the player effects from different priors and found similar plots to below, which we have in our code.



While examining the effects over each season is interesting, we also estimated a player's skill within shot type. Here we plot the top and bottom 7 players. So, over our time period, we can say that Kevin Durant had the best jumper skill, while DaQuan Jeffries had the worst dunk skill. The players with wider bars have more variability from having limited data, whereas a tighter bar means we are more certain due to having more data for them, which is also due to shrinkage from partial pooling. We see some popular players like Dončić, Durant, Curry, Nikola Jokić, Luke Kennard, Pascal Siakam, and LeBron James near the top of the leaderboards. Some players require more investigation, like Ty Jerome, who was injured in 23/24 and a role player otherwise. He did perform better than average, shooting close to 50% in our time period, though in limited opportunities. It is difficult for us to aggregate these skills across shot types as not all shots are equal. To forecast this information forward, we would need more seasons and allow the latent skill to change over time. Lastly, we can inspect the variation for the skill shot types as 0.14 for jumpers, 0.21 for layups, 0.32 for hooks, and 0.38 for dunks to understand that the dunk skill is wider, indicating more variation across players, while jumper skill has less variation.

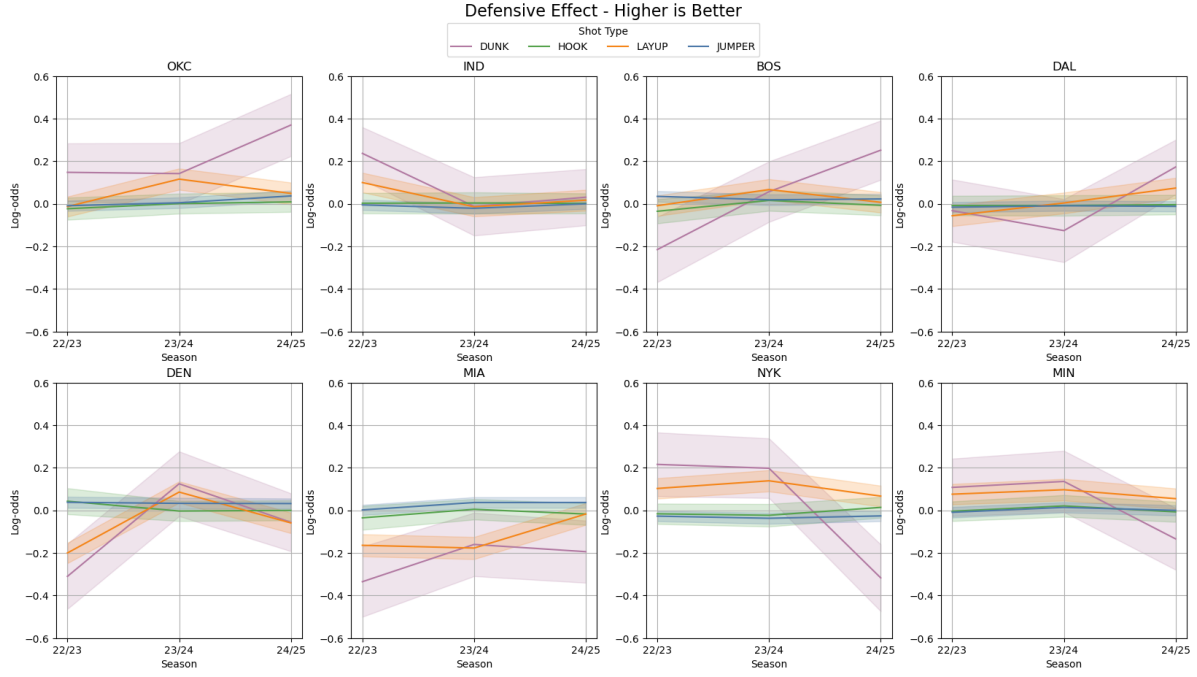
Player Skill by Shot Type - Top and Bottom 7



5.5. Defensive Effect

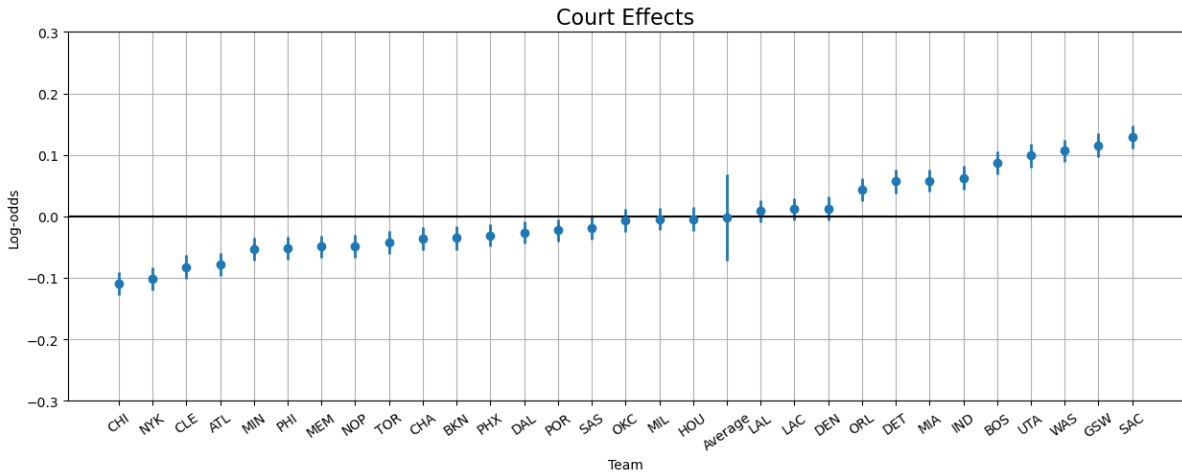
The defensive effect in our random walk model assumes each season is independent of another. We did not try the random walk for defenses due to lack of time, but we expect more variation with defenses compared to individual player ability, since defense depends on multiple players and team rosters change from year to year. Of note is the consistently small effect of team defense on jump shots. This could perhaps be caused by stronger defensive teams forcing opponents into more difficult shot locations rather than better defense against shots from the same location.

Here we show recent NBA Finals and Conference Finals teams in an attempt to understand effective defenses. In 24/25, OKC defeated IND, in 23/24, BOS defeated DAL, and in 22/23, DEN defeated MIA. We have also included MIN and NYK, the conference runners-up in the 24/25 season. The most noticeable difference between these teams is their ability to prevent dunks from being converted. Dunks only represent 5.7% of our shots and they are the most variable shot to defend given above. OKC were the best at defending dunks in 24/25, followed by MIL and BOS. OKC also had the best record overall in 24/25. In 23/24, BOS, another number one overall, were not standout in defense, but better than DAL during finals. This defensive effect does not explain everything as DAL, who were poor in defending the rim, still advanced to the Finals. We are not projecting this value forward, but merely using it to help explain why some teams are better than others. Upon inspecting the sigmas for defensive shot type, we see that most teams are pretty similar for jumpers (0.04) and hooks (0.04) with more spread in layups (0.11) and dunks (0.22). So all teams defend jumpers and hooks similarly, but there is more variance in dunk defense.



5.6. Court Effect

Here we see the court effects or court factors that show how different arenas or fan bases affect all shots equally. The Chicago Bulls, New York Knicks, and Cleveland Cavaliers all take away from shots, while shots are easier where the Washington Wizards, Golden State Warriors, and Sacramento Kings play. We did not test out tighter priors for this effect and given the size of data, it likely has little effect on the posteriors. Please see the appendix for a visual comparison of convergence for the court effect, 7.4.



6. Conclusion and Limitations

After accounting for the baseline odds of a shot given its distance and angle, we added various multi-level effects to separate player and defensive skill. We then extended this concept to estimate a player's latent ability. Our future work would look to extend this timeline to better estimate season-to-season fluctuations and transform our work into a projection model.

There were some limitations in the data that we worked with. We did not have defender identity or defender distance, which we expect would be critical factors in shot difficulty. This may have given some players a larger effect than would represent their true talent. For example, our model gave Ty Jerome one of the largest effects for jump shot, but he was a role player with only around 20 minutes per game. His team, the Cleveland Cavaliers, were one of the top teams in the league that season, so opponents may have focused their defensive efforts on more skilled teammates. Another data limitation was shot type aggregation. We lose some information by rolling up shot attempts into broad categories. A step back jump shot is different from a pullup jump shot. If we used all these categories, partial pooling would account for sample size, but some shot subtypes are only performed by a few players or in a few situations. Thus we would be estimating a trivial player skill for some players' driving reverse dunk shots or running alley oop layup shots, for example.

Besides data limitations, given BART's performance, using any frequentist model is likely sufficient. Even though modeling the geometry with BART is novel, we have not seen any public papers on it, it takes longer relative to XGBoost with only marginal benefits. A fully tuned XGBoost or Neural Network have shown to be better in this regard. Modeling shot type, player, defense, and court effects adds to the current public space because we can identify skill after accounting for various attributes. Further, modeling player skill over time should not assume independence year over year. While the random walk accomplishes some of this, extending this concept to allow for correlation between shot type effects for players would be interesting, as proficiency at scoring layups could be correlated with ability to make dunks or even other shot types. We hypothesize that one could measure this through the use of a multivariate random walk. With this approach, one could draw innovations from the correlated walk for each shot type and gain understanding of a player's skill evolution and shot type correlations. Secondly, our model is not a projection model, and is instead more inference. In sports, this makes the model limited because teams focus on forecasting performance to understand how to win. Teams are likely using more advanced structures to predict performance. With our approach and more seasons, we can bridge the gap between the frequentist public models and forecasting ones.

7. Appendix

7.1. Shot Type Mapping

Shot Type	Mapped subType values
LAYUP	Tip Layup Shot, Driving Layup Shot, Running Layup Shot, Cutting Layup Shot, Driving Reverse Layup Shot, Layup Shot, Driving Finger Roll Layup Shot, Running Reverse Layup Shot, Running Finger Roll Layup Shot, Alley Oop Layup Shot, Putback Layup Shot, Reverse Layup Shot, Finger Roll Layup Shot, Cutting Finger Roll Layup Shot, Running Alley Oop Layup Shot
JUMPER	Jump Shot, Driving Floating Bank Jump Shot, Driving Floating Jump Shot, Fadeaway Jump Shot, Step Back Jump Shot, Running Jump Shot, Pullup Jump Shot, Floating Jump Shot, Running Pull-Up Jump Shot, Turnaround Fadeaway Bank Jump Shot, Jump Bank Shot, Turnaround Jump Shot, Step Back Bank Jump Shot, Turnaround Fadeaway Shot, Turnaround Bank Shot, Fadeaway Bank Shot
DUNK	Running Dunk Shot, Cutting Dunk Shot, Driving Dunk Shot, Putback Dunk Shot, Running Alley Oop Dunk Shot, Tip Dunk Shot, Dunk Shot, Reverse Dunk Shot, Alley Oop Dunk Shot, Running Reverse Dunk Shot, Driving Reverse Dunk Shot
HOOK	Hook Shot, Turnaround Bank Hook Shot, Turnaround Hook Shot, Driving Hook Shot, Hook Bank Shot, Driving Bank Hook Shot

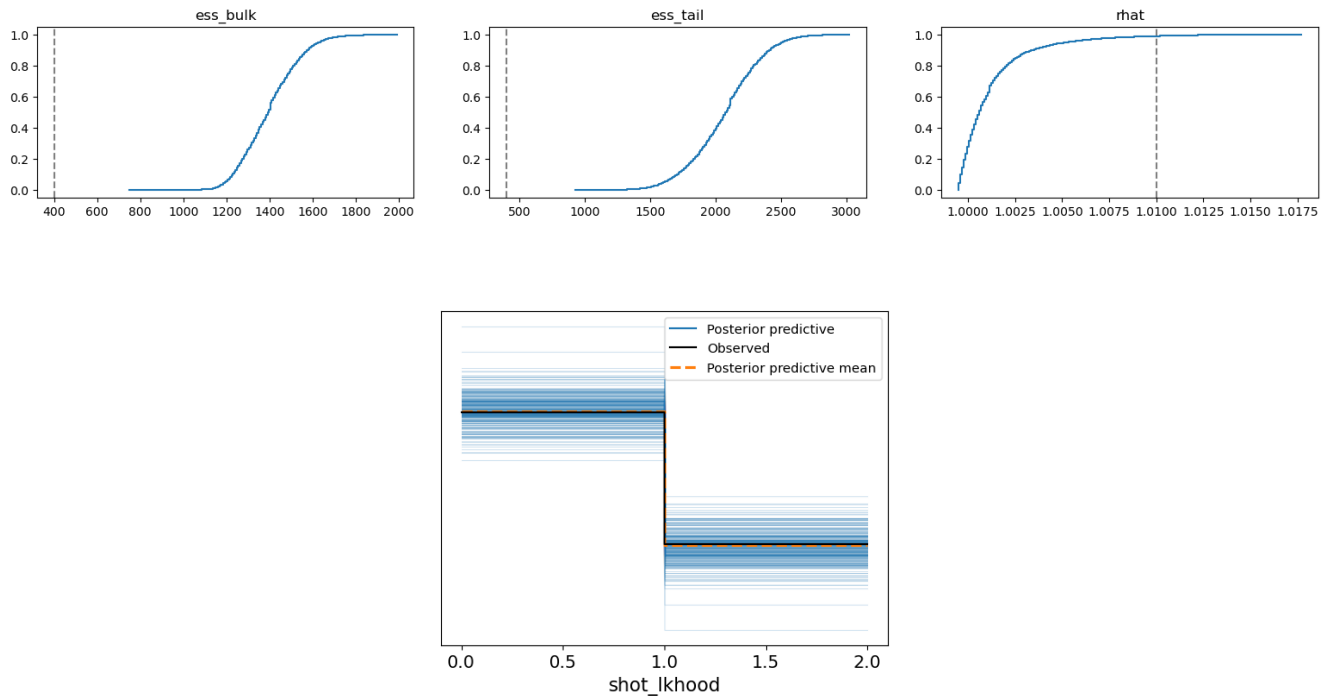
7.2. Performance Review

McFarlane used XGBoost on distance, angle, and shot type with data from 2018. He trained two models, one on NBA and another on WNBA, with cross validation and reported a LogLoss of 0.641 and 0.652 respectively. While LogLoss is not an apples to apples metric with a different dataset, it is in the ballpark of our performance. Cen et. al used a Logistic Regression with Elastic Net penalty and achieved a 65% accuracy. They also used features such as defensive distance, coordinate location, shot clock features, player height, defender height to name a few. Murakami-Moses examines a Logistic Regression, Neural Network and XGBoost and achieves an accuracy between 64.9%-65.1% across the three models. This paper also used features such as clock, shot zone area, shot zone basic, and coordinate location. Meehan analyzed the most number of models from Support Vector Machines to Neural Networks to Naive Bayes. These models ranged from 54%-68% accuracy. Their features were similar to Murakami-Moses. It is disappointing the majority of these models only report a single metric and focus solely on accuracy. Secondly, with our baseline model performance on par with these public models, it highlights that additional features beyond distance and angle are not providing substantial improvements to predictive power. We do advocate for the use of shot type similar to McFarlane’s work, although given our two stage approach, swapping BART with a different geometric approach provides future flexibility.

7.3. BART Convergence

We first examine how BART estimates the whole function given the ESS and R-hat. The first plot shows the empirical CDF of those values. We want the ESS values to be above the dashed line and the for R-hat we want the entire curve to be below the dashed line. In the second plot,

we look for the overlap between the orange and the black line to confirm the posterior predictive distribution aligns with our observed data.



7.4. Parameter Convergence

